

Dressed normal moments on symmetric boxes: asymptotics and parity

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Abstract

Second and final unit of the signed-reflection extension. Combining the exact reduction of unit 196 with the continuous-amplitude theorem of unit 186: the symmetric-box dressed normal moment, normalised by $N^{-\lambda}(\log N)^{|J|-1}$, converges to the face coefficient of the signed symmetrisation $\eta_{\text{sym}}(\text{symmetric_amplitude_tendsto})$; for equal ratios the limit is $\eta(0) \prod_i (1 + (-1)^{h_i}) \cdot \Gamma(\lambda) \beta^{-\lambda} / (m! \prod_i 2k_i)$ ($\text{symmetric_amplitude_equal}$), so **an odd exponent annihilates the equal-ratio leading corner coefficient** ($\text{symmetric_amplitude_odd}$): the integral is then $o(N^{-\lambda}(\log N)^m)$; it does not vanish, and no next-order rate is claimed. Headline XI wrappers restate this on the Bochner integrals. Zero `sorry/axiom`.

1 The things formalised

```
theorem symmetric_amplitude_tendsto (d) (h k : Fin (d+1) → ) (hk) (l ) (hl) (h )
  (hmin) (hatt) ( ) (h : Continuous ) :
  Tendsto (fun N => ( x in symBox (d + 1), x * (( i, x i ^ h i)
    * exp (-( * N * i, x i ^ (2 * k i))))
    / (N ^ (-1) * log N ^ (multCount (ratioExp h k) l - 1))) atTop
    ( (amplitudeCoeff h k l (symAmp h )))
theorem symmetric_amplitude_equal ... (hratio : i, ratioExp h k i = 1) :
  Tendsto (... / (N ^ (-1) * log N ^ m)) atTop
    ( ( 0 * ( i, (1 + (-1 : ) ^ h i))
      * (Gamma l * ^ (-1) / (m.factorial * i, 2 * k i))))
theorem symmetric_amplitude_odd ... (j) (hodd : Odd (h j)) :
  Tendsto (... / (N ^ (-1) * log N ^ m)) atTop ( 0)
theorem headline_symmetric_normal_moment_amplitude
theorem headline_symmetric_normal_moment_equal
```

2 Paper-facing meaning

Interior normal coordinates with a continuous amplitude are now covered: the leading functional is the face functional of the signed symmetrisation, and the paper's parity mechanism appears as the factor $\prod_i (1 + (-1)^{h_i})$ on the corner value in the equal-ratio case. Astra #18's caution is formalised exactly: an odd exponent kills the corner coefficient (hence little- o of the bare scale), not the integral, and no faster rate follows.

3 Lean notes

Both theorems are one-line compositions of `integral_symBox_eq_symAmp`, `amplitude_tendsto`, `amplitudeCoeff_equal` and `symAmp_zero`; the odd case is `Finset.prod_eq_zero` with `Odd.neg_one_pow`. The only friction was the usual beta-redex in a `Tendsto.congr'` goal.